LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Thrust augmentation and Thrust augmentation by water injection and water methanol injection system After burning operation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Thrust augmentation and Thrust augmentation by water injection and water methanol injection system After burning operation
5. Specific Objectives	13.1.3.1 Trainee will be able explain methods of thrust augmentation

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT slides on thrust augmentation cycles; Instructor supervision of trainee note-taking on system components and operational limits.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of thrust augmentation?
(b) Name one method used to increase engine thrust beyond maximum continuous rating.
(c) How does afterburning affect the exhaust gas temperature?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of thrust augmentation • Thermodynamic principles of afterburning • Water injection system components and operation • Water methanol injection system advantages • Operational limitations and safety constraints	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of water injection cooling effects • Comparison of water vs water methanol injection • Impact on specific fuel consumption and thrust • System integration with engine control units • Maintenance requirements for injection systems	L & Ds	PPT, PDF	20 min
(c)	• Afterburning operation sequence and activation • Thermal protection measures in exhaust nozzles • Fuel flow management during augmentation • Emergency shutdown procedures for injection systems • Performance gains in supersonic flight regimes	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of water injection in reducing exhaust gas temperature.
(b) Differentiate between water injection and water methanol injection systems.
(c) Describe the operational sequence for activating afterburning.

12. Home Assignment and Follow Up	Prepare a brief note on how thrust augmentation techniques will be utilized in the next lesson regarding noise suppression and its levels.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Noise suppression and its levels	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Verify specific reference book editions for water methanol injection system diagrams.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min